

```
# cp /home/sandeep/Fedora-9-i386-DVD.iso /fedora/fedora9.iso
```

Now extract the kernel and initrd files from the ISO image and place them in the same directory in which you placed the ISO. You can use File Roller, the archive manager for GNOME, to extract the files. Just right click on the ISO and select *Open with File Roller*. It displays the contents of the ISO image. Then navigate to the *isolinux* directory—in Fedora 9 these two files are placed inside the *isolinux* directory; it's often different for other distros, so please refer to Table 1 for the paths. Select the kernel and initrd files, and extract them to the location where your ISO image exists.

The second method is to mount the ISO image and extract the files. Run the following commands to do this:

```
# mount -o loop /fedora/fedora9.iso /media/iso
# cd /media/iso/isolinux
# cp vmlinuz initrd.img /fedora/
```

I have mounted the ISO image without providing the *-t iso9660* option (to specify the type of media as an ISO filesystem). It worked for me. If the above mount command doesn't work, do add this option along with the rest of the *mount* command above.

Note: Fedora 10 has introduced a change in the Anaconda installer. So, in addition to the *vmlinuz* and *initrd.gz* files, you will also need to copy the *images/install.img* file, create a directory called */fedora/images*, and place the *install.img* file there.

Now, it's time to edit the */boot/grub/menu.lst* file on the system I'm currently using—Ubuntu 8.10. Note that this is the location of the Grub menu in almost all distros, except for Fedora/Red Hat, where it's called */boot/grub/grub.conf*. Append the following entry there:

```
title Install Linux
root (hdX,Y)
kernel /distro/Linux_kernel
initrd /distro/Ram_disk
```

In this case...

- 'title' is the name you want to display in your GRUB menu
- 'root' is the hard disk partition that contains the ISO image
- 'kernel' is the Linux kernel
- 'initrd' is the initial RAM disk image

Likewise, the *menu.lst* entry for the ISO file looks like what's shown below:

```
title Install Fedora 9
root (hd4,0)
kernel /fedora/vmlinuz
initrd /fedora/initrd.img
```

Names of kernel and RAM disk images in some popular distros

Linux OS	Kernel path	Ram disk path
Fedora	/isolinux/vmlinuz	/isolinux/initrd.img
openSUSE	/boot/386/loader/linux	/boot/386/loader/initrd
Mandriva	/i586/isolinux/alt0/vmlinuz	/i586/isolinux/alt0/all.rdz
Ubuntu	/casper/vmlinuz	/casper/initrd.gz
Debian	/install.386/vmlinuz	/install.386/initrd.gz
RHEL5/CentOS5	/isolinux/vmlinuz	/isolinux/initrd.img

Table 1

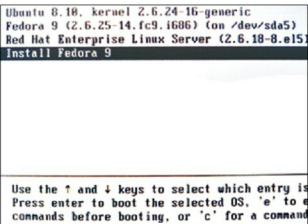


Figure 1: The Install Fedora 9 Grub menu entry

Now you are ready to install your new Linux distro directly from the hard disk without the need for a CD/DVD drive. Reboot your system and select the 'Install Fedora 9' entry from your GRUB menu.

Figure 1 shows what the GRUB menu looks like after rebooting my system.

Obviously, I selected the 'Install Fedora 9' entry and it has started booting my system with the help of *vmlinuz* and *initrd.img* files. The set-up prompts me to choose a language and keyboard layout. Then it prompts me to select the 'Installation Method' as shown in Figure 2.

In this screen you need to select the 'Hard drive' option and proceed to the next screen. Here, you have to select the appropriate partition and the directory where the installation image exists. In my system, the installation image exists in the */fedora* directory of */dev/sda5* partition. This is shown in Figure 3.

...continued on page 27



Figure 2: Select the hard drive for 'Installation Method'